

Evidence-Indexed Slide Narration With Heuristic Checks and Template Fallback

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Abstract

Slide narration generation can introduce unsupported visual or decorative claims. We describe a pipeline that indexes slide content with deterministic identifiers, generates an internally cited template, and applies heuristic checks before and after an optional language-model rewrite. A narration-stage study covers 30 publicly sourced presentations (279 slides). The same decks and procedure were used, but stochastic vision evidence was regenerated by condition. In one run per condition, the complete configuration produced 5/279 proxy flags (1.79%) versus 22/279 (7.89%) for A1, a 77.3% relative difference (slide-paired exact $p \approx 2.2 \times 10^{-4}$; deck-blocked $p = 0.0156$); B produced 20/279 (7.17%). Saved labels mark 225/279 condition-C outputs as template fallbacks. Post-rewrite verdicts were not saved, and six outputs retained model narration when no nonempty verified template was available. The proxy has $F1 = 0.60$ against judgments based on extracted context rather than rendered slides. These are observed system-level proxy rates, not component attribution or evidence that hallucination is eliminated.

1 Introduction

Slide decks combine running text, shape labels, connectors, embedded images, screenshots, and speaker notes. When a language model is asked to narrate a slide, it can fill in details that are nowhere in the source: a chart trend that is not in the data, a participant in a diagram that does not exist, or decorative imagery (“a vibrant landscape”, “a serene backdrop”) for a slide that only contains a logo and a bullet list. In narrated video, these claims become part of the audio track and the subtitle file, and they cannot easily be retracted.

Putting the slide content into the prompt is the obvious first step, and it is what most slide-narration tools already do. In the conditions we report below, explicit evidence fields alone do not

yield a noticeable change from the first configuration in proxy-flagged unsupported claims. The model still adds visual descriptions about images it cannot see and inserts decorative adjectives that no part of the slide supports.

This paper describes a working system that takes a different route. We treat the template script as evidence-indexed claim generation. Each template segment carries stable identifiers into an evidence index built from the slide contents. A deterministic verifier checks citation integrity and other heuristic properties before the script leaves that stage. After the optional language-model rewrite, a second check evaluates the rewritten slide against the combined evidence scope of its surviving template segments. A failed rewrite is replaced when a nonempty template is available. Six preserved condition-C slides had no surviving template segment and retained model text: this is a fail-open edge case, not verified fallback. The application supports slide-to-video execution, but the formal experiment used `--skip-av` and evaluated outputs through the narration stage.

The contributions are:

1. **A deterministic evidence index and citation contract:** every template-script segment must cite SHA-256 identifiers that point at extracted slide content. Identifiers are stable within a pipeline run.
2. **A closed-loop heuristic verifier** with 9 checks (image-claim grounding, evidence-existence, graph-consistency, hedging on low-confidence vision evidence, and others), which iterates with the script generator until either every claim passes or the iteration budget is reached.
3. **Post-rewrite checking with conditional template fallback:** after the language-model rewrite, a slide-level check runs against the

union of the surviving template segments’ identifiers. Failed rewrites are replaced only when a nonempty fallback target exists.

4. **A narration-stage evaluation** on 30 publicly sourced decks (279 slides) showing an observed 77.3% relative difference in proxy-flagged unsupported narration for the complete system configuration ($p \approx 2.2 \times 10^{-4}$ slide-paired; $p = 0.0156$ deck-blocked) compared with the configuration that omits explicit evidence fields and initial verification.
5. **An extracted-context annotation set** (100 manually labeled records) and a discussion of the proxy, annotation-view, and study limits.

The paper is intentionally narrow. We are not claiming a solution to hallucination. We report a lower observed proxy-flagged rate for one system configuration on a real corpus, and we are being explicit about what that proxy does and does not capture.

2 System Overview

The system processes one PPTX deck at a time through a fixed sequence of stages, shown in Figure 1. Only the stages that are load-bearing for the grounding mechanism are discussed here; downstream rendering, audio, and video composition are sketched only to the extent that they shape what can and cannot be claimed in narration.

2.1 Input and evidence extraction

The deck is parsed with `python-pptx`. We pull out every text span (running text, titles, body, list items), every shape label, every connector, and every embedded image. Speaker notes are kept verbatim. For slides with embedded images, we also call a vision model (GPT-4o-mini) to produce captions, object tags, and a one-line diagram summary. Native PPT evidence (text, shapes, connectors) is given confidence 1.0; vision evidence is given a confidence score read from the vision response. For a claim, the verifier uses the minimum confidence among its cited image evidence and requires hedging below the configured threshold.

2.2 Evidence identifiers and provenance

Each extracted item is assigned a stable identifier

$$\text{eid} = \text{SHA-256}(\text{job_id} \parallel \text{slide_idx} \parallel \text{kind} \parallel \text{offset})$$

where `kind` is one of about twenty evidence kinds and `offset` disambiguates multiple items of the same kind on the same slide. Identifiers are deterministic within a pipeline run, which is sufficient for downstream verification and for the human-readable evidence report that the system emits alongside the video. We do not claim cross-run stability; that property is not needed and would complicate concurrent runs sharing storage. The kinds used in this work are ten native kinds (`TEXT_SPAN`, `SHAPE_LABEL`, `CONNECTOR`, `IMAGE_ASSET`, `TABLE_HEADER`, `TABLE_CELL`, `CHART_TITLE`, `CHART_CATEGORY`, `CHART_SERIES`, `CHART_VALUE`) and a set of vision-side kinds (`IMAGE_CAPTION`, `IMAGE_OBJECTS`, `IMAGE_ACTIONS`, `IMAGE_TAGS`, `DIAGRAM_TYPE`, `DIAGRAM_ENTITIES`, `DIAGRAM_INTERACTIONS`, `DIAGRAM_SUMMARY`, `SLIDE_CAPTION`). Native PowerPoint tables are walked cell by cell. When chart data is available either as a native PPT chart (Insert $\rightarrow$ Chart) or as an embedded Excel OLE workbook, the extractor walks the chart structure and emits `CHART_TITLE`, `CHART_CATEGORY`, `CHART_SERIES`, and one `CHART_VALUE` item per (*series*, *category*) data point (e.g., Win2k Service Packs / SP1 = 71). Screenshot-only charts still fall back to vision captioning. The `TABLE_*` and `CHART_*` kinds are part of the current artifact, but were added after the headline experiment in Section 4.4; their effect on the reported numbers is discussed in Section 5.2 and Limitations.

2.3 Narration generation with citations

The script generator emits a list of narration segments. Each segment has a slide index, text, a list of `evidence_ids` the segment is supposed to be grounded in, and a list of `entity_ids` pointing into the slide’s knowledge graph. The default script generator in the evaluated system is a deterministic template that walks the evidence index slide by slide; a language-model script generator is available as an alternative provider. These identifiers belong to the internal script artifact. The final slide-level text sent to speech does not retain a literal evidence-ID list for each rewritten sentence.

2.4 Closed-loop verification

Described in detail in the next section.

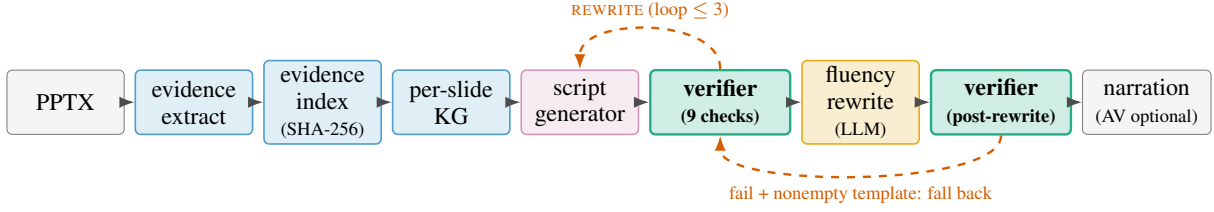


Figure 1: Pipeline. **Blue**: evidence extraction. **Purple**: script generation. **Green**: the two heuristic-checking stages. **Orange**: the optional LLM fluency rewrite. The first verifier loops back to the script generator (top arrow); after the second check, failure causes fallback only when a nonempty template exists.

2.5 Translation, subtitles, speech, and video

After the verifier, the verified script is optionally translated. The system re-verifies the translated narration for citation presence and graph consistency (function `verify_translated_script` in the codebase), but content-level paraphrase validation is not redone after translation. Subtitles are emitted sentence by sentence with WebVTT and SRT. Speech is synthesized with one of the configured TTS providers (system voices on macOS, an HTTP-based provider otherwise). Video is composed with FFmpeg from the rendered slide PNGs and the per-slide audio tracks. None of these stages is the focus of this paper.

3 Closed-Loop Verification

The verifier is the technical center of the grounding mechanism. Its pre-rewrite unit is one template segment, which is roughly one sentence. It is a deterministic rule system, not a learned semantic-entailment model.

3.1 Checks

The verifier runs nine checks per segment. They are listed in Figure 2. The checks cover citation presence, citation validity, entity-ID grounding in the slide graph, image-claim grounding against image evidence kinds, object/action consistency, diagram-interaction consistency, token-level support against cited evidence, hedging when the cited evidence has low confidence, and relation consistency against the graph. The content-support rule enforces numeric consistency and requires at least one token shared with the cited evidence; it does not establish semantic entailment. We therefore treat PASS as “passes these checks,” not as a human-validated guarantee of factual support.

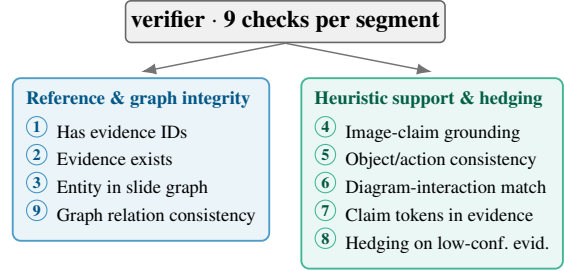


Figure 2: The verifier’s nine checks per narration segment, clustered. **Left**: structural integrity of citations and graph references (uncited segments, stale IDs, references to nodes that do not exist on the slide). **Right**: heuristic checks on claim content and cited evidence, with hedging required when cited evidence is low-confidence vision.

3.2 Verdicts

The evaluated implementation emits two verdicts in practice: PASS and REWRITE. A third verdict, REMOVE, is defined in the interface but is not emitted by the current decision logic in our evaluated runs (the per-condition aggregate for the full pipeline shows zero REMOVE verdicts; see Table 4). We mention this explicitly so that the verdict surface in the paper matches what the code actually does in this setting.

3.3 Rewrite loop

When the verifier returns REWRITE, the offending segment is regenerated. In the evaluated deterministic script generator the rewrite uses templated hedging (e.g., “shows” becomes “appears to show” if the cited evidence is low-confidence vision evidence). The loop runs up to three iterations.

3.4 Post-rewrite verification

After the optional language-model fluency rewrite, the verifier runs once more on the entire rewritten slide. The implementation constructs one synthetic segment using the deduplicated union of

evidence and entity IDs from its surviving template segments. When this segment does not pass and the template is nonempty, the rewrite is replaced by that template. The artifacts do not persist the post-rewrite verdict, only selected text and a `source_used` label; that label is not a pass/fail result. Six C slides had no surviving template segment. Reconstructed historical code gives those slides an empty-ID check that cannot pass, but retains model text because no fallback text exists. Section 4.4 separates this exception from the output-source counts.

4 Evaluation

4.1 Corpus and experiment

The working corpus contains 157 presentations (4,009 slides; median 20 slides per deck) collected from publicly accessible U.S. government and government-related websites. Preserved project metadata associates the corpus with Library of Congress digital collections (Library of Congress, n.d.), but we do not claim that every source deck is public domain. The formal experiment uses a fixed 30-deck subset (279 slides) listed in the supplementary material. The release supports recomputation of reported statistics from saved outputs; it does not contain source deck binaries or support an exact experiment rerun. Conditions A1, B, and C operate on the same 30 files and the same 279 slides.

4.2 Conditions

All three conditions use the same deck files, slides, extraction and rendering procedure, slide text and notes, deterministic-template procedure, rewrite LLM (GPT-4o-mini, temperature 0.7, max_tokens 300), and post-rewrite-checking code. Stage caching was disabled, so vision-derived evidence was regenerated independently rather than held fixed across conditions. A1 omits the prompt’s explicit EVIDENCE and KEY ELEMENTS fields and skips initial verification. B adds those explicit fields but still skips initial verification. C adds the initial verifier and uses surviving verified-template text as fallback when nonempty; A1 and B use unverified templates. Thus A1 is not a raw baseline, C changes multiple components, and condition differences include stochastic vision-extraction variation. No raw or template-only condition was evaluated.

Cond.	Evid.	Init. check	Fallback	Flag.	Rate
A1 (no explicit evid.)	—	—	unverif.	22/279	7.89%
B (explicit evid.)	✓	—	unverif.	20/279	7.17%
C (full pipeline)	✓	✓	verified*	5/279	1.79%

Table 1: Conditions and main result. Every prompt includes an evidence-derived template, slide text, and notes; “explicit evidence” means the additional evidence and graph fields. Vision evidence was regenerated per condition. *C uses a verified template only when nonempty; six C slides lacked one and retained model text. A1 vs. C exact $p \approx 2.2 \times 10^{-4}$; B is not distinguishable from A1.

4.3 Metrics

We measure the proxy-flagged-unsupported-narration rate per slide. A slide is flagged if its narration text matches any of a 24-word visual-embellishment vocabulary (*vibrant, serene, intricate, stunning, landscape, rural, scenic, colorful, abstract, artistic, microscopic, aerial, panoramic, beautiful, elegant, captures, gazing, glimpse, vivid, bright, lush, backdrop, swaying, tranquil*; matching is case-insensitive substring matching, which can over-trigger on inflected forms such as `lush` inside `flush`). The full vocabulary, including the 15-word image-claim vocabulary used elsewhere in the codebase, is in the supplementary material and in the released evaluator (`code/proxy_evaluator.py`). The metric is the fraction of slides flagged by at least one keyword.

We calibrate the proxy against 100 manually labeled extracted-context records drawn across conditions. Annotators saw slide text, shape labels, model-generated image captions, and narration, but not the PowerPoint or rendered slide; some audit fields were clipped at about 200 characters. The labels therefore measure support relative to supplied extracted context. We treat the proxy as a partial, moderate-fidelity stand-in. It is sensitive to descriptive visual embellishment (“a panoramic backdrop”) and insensitive to other hallucinations that avoid the vocabulary (“improving patient outcomes” on a slide that does not say that).

Significance is reported as a paired permutation test on the per-slide binary labels of conditions A1 and C, with random sign flipping of the per-slide differences. For binary paired data this is equivalent to McNemar’s exact test on the discordant pairs. With 19 slides flagged in A1 but not C and 2 flagged in C but not A1 (21 discordant pairs total, 3 concordant flagged in both, 255 concordant

clean), the exact two-sided $p \approx 2.2 \times 10^{-4}$. A Monte Carlo permutation test with $n = 100,000$ shuffles and seed 42 gives $p \approx 2.6 \times 10^{-4}$, consistent with the exact value. The script that computes the test is released at `code/compute_permutation_test.py`; the exact analytical p-value can be reproduced from the discordant-pair counts above. Because slides within one deck are related, we also sign-flip all paired slide differences within a deck together. This exact deck-blocked sensitivity test gives $p = 0.015625$. Two decks account for 10 of the net 17 fewer flags, so the slide-level result should not be read as 279 independent deck-level replications.

4.4 Results

Table 1 reports the preserved run. The observed proxy-flagged rate differs from 7.89% in A1 to 1.79% in C, a 77.3% relative difference. Condition B (evidence in the prompt only) is 7.17%, not distinguishable from A1 (exact paired $p = 0.5$). Because the keyword proxy has $F1 = 0.60$ against one annotator’s extracted-context labels (Table 2), this result should be read as an observed difference in one class of proxy-flagged language, not a complete measurement of hallucination. It is a configuration-level association: the design neither isolates a component nor holds stochastic vision evidence fixed.

Table 2 reports calibration on a separately exported 100-record set using the same condition labels and input decks. Reviewers labeled supplied extracted context without inspecting original slides. The proxy reaches $F1 = 0.60$ against one annotator’s binary labels on 100 extracted-context records (precision 0.65, recall 0.57). A second human annotator independently re-labeled the same 100 records using a 4-way schema (supported / partially_supported / unsupported / unclear); inter-annotator agreement was Cohen’s $\kappa = 0.54$ on the 4-way schema and $\kappa = 0.51$ on the binary collapse (`{partially, unsupported}` $\rightarrow$ flagged). A targeted manual audit of 28 condition-C examples in that annotation set (Figure 3) gives precision 0.75, recall 0.55, $F1 = 0.63$. The exported narrations are not byte-identical to the preserved final JSON used for the 1.79% rate. This is extracted-context calibration on related outputs, not direct slide inspection or relabeling of the 279 final outputs.

Table 3 reports final source-label counts. A `post_rewrite_fallback` label records selected output, not a saved post-rewrite verdict. In

	proxy=YES (n=8)	proxy=NO (n=20)	
fabricated	4	5	<i>missed</i>
embellishment	2	0	
metaphor_fp	1	0	<i>false alarm</i>
clean	1	15	

P = 0.75 R = 0.55 F1 = 0.63 (binary collapse)

Figure 3: Manual audit of 28 condition-C examples in the separately exported annotation set (8 proxy-flagged + 20 randomly-sampled unflagged). Rows: 4-way audit labels; columns: proxy decision. Green = true negatives; vermilion (bold cell) = the 5 context-relative unsupported additions the proxy missed (no flowery vocabulary); blue = metaphor or clean records the proxy flagged. The audit converts the $F1 = 0.60$ weakness into a concrete failure mode: the proxy catches embellishment but misses semantic unsupported additions, so the headline 1.79% is best read as a narrow proxy rate rather than as elimination of hallucination.

	TP	FP	FN	TN	P	R
Proxy vs. manual	26	14	20	40	0.65	0.57

Table 2: Proxy calibration against reviewer-1 labels on 100 extracted-context records. $F1 = 0.60$. Annotators did not inspect the original rendered slides.

C, 225/279 entries use verified-template text and 54 are labeled `ai_narrate`. Six of those 54 have no passing template segment in the saved initial report. Reconstructed code gives their synthetic check an empty evidence-ID union and retains AI text because the fallback is empty. We report an 80.6% fallback-selection rate, not that all 54 rewrites passed.

Table 4 reports verifier verdict distribution for condition C reconstructed from the on-disk per-slide `verify_report.json` artifacts. Aggregated across the 30 decks, the verifier emitted 510 PASS, 1499 REWRITE, and 0 REMOVE verdicts on the script before fluency rewrite.

Only three reason codes fired in the condition-C reports: `IMAGE_UNGROUNDED` (1,471), `NO_EVIDENCE_IDS` (1,451), and `OBJECT_ACTION_UNSUPPORTED` (28); one segment can carry multiple reasons. The supplementary contains a real A1–C narration pair and a schematic, explicitly non-empirical explanation of the internal identifier contract.

5 Operational Lessons for NLP Systems

The broader application is a slide-to-video system that can emit video, subtitles, an evidence report,

Condition	Slides	Fallback	Rate
A1: no explicit evidence	279	245	87.8%
B: explicit evidence	279	244	87.5%
C: full pipeline	279	225	80.6%

Table 3: Final source labels, not post-rewrite pass/fail counts because those verdicts were not saved. Six C outputs expose an empty-template fail-open case.

Verdict	Count	Share
PASS	510	25.4%
REWRITE	1499	74.6%
REMOVE	0	0.0%

Table 4: Verifier verdicts on 2,009 claim segments across the 30 condition-C runs. The high REWRITE share is a consequence of the deterministic template generator preferring to emit one claim per extracted item; rewriting moves text toward cited evidence or adds hedging, and terminal nonpassing segments are omitted.

and a review queue. The formal 90-run experiment used `--skip-av`; audio and video were skipped and every saved `output_video` is null. The evaluated path ends at narration and diagnostics. The following implementation telemetry is not an AV evaluation.

5.1 Operational grounding gate

At run time, verifier output and narration-source bookkeeping are converted into a single delivery decision per deck: `DELIVER`, `DELIVER WITH WARNINGS`, `NEEDS REVIEW`, or `BLOCK`. The gate combines six signals: missing citations and invalid evidence IDs from the verifier; the fallback rate from the narration source log; the proxy-flagged unsupported-narration rate; low-confidence vision claims actually used in narration; chart-or-table risk on slides narrating tabular content without native table evidence kinds; and translation warnings on non-English variants. This is what makes the grounding signals available to a delivery workflow rather than only an offline score; we do not claim that it reduces reviewer time.

Signal semantics. The `missing_citation_count` signal counts script segments where the verifier emitted `NO_EVIDENCE_IDS` after the loop converged. It reflects script-time grounding state, not the post-rewrite final narration (which may select template fallback). `invalid_evidence_id_count` is defined analogously on `EVIDENCE_NOT_FOUND`.

Profile	Deliver	Warn	Review	Block
strict audit	0	0	0	30
production review	0	0	30	0

Table 5: Gate decisions on the 30 condition-C decks. The strict profile blocks all decks because each has at least one converged `NO_EVIDENCE_IDS` flag; the production profile routes them to review.

Two threshold profiles. The *strict audit* profile blocks on any missing-citation flag, any invalid evidence ID, or a proxy-flagged rate above 10%; it is intended for offline evaluation and CI checks. The *production review* profile is the configured default: missing-citation and invalid-evidence-id flags route to review, and only an extreme proxy rate (above 25%) blocks outright. Both profiles route fallback, low-confidence-vision, chart/table, and translation-warning signals to `NEEDS REVIEW`. Table 5 reports gate decisions on the 30 condition-C decks under each profile.

Cost and latency. Render and vision captioning dominate measured wall-clock (5/9/15-slide buckets, $n = 3/5/2$: 19.95/39.26/92.58 s; render plus vision >80%). The historical artifacts do not record token usage, so we cannot report measured model cost.

Fallback behavior. Source labels mark 80.6% of C slides as using verified deterministic-template text. This is selection, not a post-check failure rate; verdicts were not saved and six AI-labeled outputs had no nonempty verified target. Fluency, coverage, redundancy, and preference were not measured.

5.2 Reviewer calibration and review-queue usefulness

Reviewer calibration. The supplementary includes a four-way schema, 100 extracted-context records, and agreement code. Annotators saw extracted text, shapes, generated captions, and narration, not the PowerPoint or rendered slide. Independent agreement is $\kappa = 0.54$ (four-way), $\kappa = 0.51$ (binary collapse), and 76% raw; these values are context-view limited.

Review-queue usefulness. Across the 30 condition-C decks (2,009 claim segments), the gate emits 3,273 review items, a mean of 109 per deck (median 81, nearest-rank p95 229). Multiple risk reasons can produce items for one segment, and every slide appears in the queue; all 30 decks require full manual review under these pro-

files. These are diagnostic telemetry, not evidence of reduced review effort. The per-deck CSV is in the supplementary.

6 Related Work

Faithful summarization, knowledge-grounded dialogue, and RAG condition generation on external content (Maynez et al., 2020; Dinan et al., 2019; Lewis et al., 2020). Our setting differs because the evidence is a closed, per-deck index with exact identifiers rather than retrieved passages. Post-hoc consistency checks, hallucination surveys, and FEVER-style verification study factuality after generation (Manakul et al., 2023; Ji et al., 2023; Thorne et al., 2018). Our verifier instead runs at claim level before and after rewrite, and its outputs drive fallback and review routing. We do not compare against an external slide-narration system because no common benchmark or compatible public output set was used in this study; A1, B, and C are internal system configurations.

Limitations

The fabrication proxy is keyword based. F1 against one annotator’s labels on 100 extracted-context records is 0.60. The proxy will miss unsupported claims that avoid its 24-word vocabulary (e.g., invented inferences such as “improving patient outcomes” on a slide that does not say that) and it may flag legitimate uses of those words. We read the headline result as an observed lower proxy-flagged rate in C than A1, not as elimination of hallucination or an isolated component effect.

A targeted manual audit of condition-C examples makes this concrete (Figure 3). The 100-record manual set is a separately exported annotation set stratified across conditions for proxy validation, so it contains a higher density of flagged records than the 279-slide formal experiment (8 of 29 C slides flagged in the manual set, vs. 5 of 279 in the formal experiment). We inspected all 8 proxy-flagged condition-C slides plus 20 randomly-sampled unflagged condition-C slides, giving precision 0.75, recall 0.55, $F1 = 0.63$ on the stratified subset. The dominant failure mode is missing context-relative unsupported additions that avoid the 24-word vocabulary (5 of 20 unflagged records contained invented entities or numbers without flowery wording); metaphor over-flagging is rare (1 of 28 cases). The audit CSV is in the supplementary. These annotation

narrations do not byte-match the preserved final narration JSON used for the headline rates, so the audit calibrates the proxy on related study outputs rather than directly relabeling the 279 preserved final outputs.

The manual validation is anchored on one annotator’s binary labels; the proxy $F1 = 0.60$ is against that reference. A second-annotator pass on the same 100 records using a 4-way schema gives Cohen’s $\kappa = 0.54$ (4-way) / 0.51 (binary collapse). This indicates moderate agreement, and the main disagreement was the supported / partially-supported boundary, reflecting genuine task ambiguity on slides whose narration adds minor paraphrastic detail. A larger multi-annotator study would tighten this further.

The A1–C comparison changes both initial verification and the fallback target. It therefore evaluates complete configurations rather than the causal contribution of one component. A truly raw baseline and a verified template-only baseline were not run. The verifier itself was not directly validated against human claim-level judgments, so false-pass and false-rewrite rates for its nine checks remain unknown.

Post-rewrite verdicts were not persisted, so `source_used` supports selection counts but not a claim that every retained rewrite passed. Six C slides had no surviving verified template: the reconstructed checker rejects their empty identifier union, but an empty fallback leaves AI text in place. The historical system was not fail-closed on those slides.

The primary test treats slides as paired units although slides cluster within decks. Deck-blocked sensitivity remains significant ($p = 0.015625$), but two decks contribute 10 of the net 17 fewer flags. Stage caching was disabled and vision evidence regenerated by condition: normalized A–C artifacts had 1,721 common, 446 A-only, and 447 C-only non-asset evidence records; all differences were vision-derived. With one realization per condition, vision and rewrite variance are not estimated.

The 30-deck experiment is a subset of the 157-deck corpus. The 30 files were selected once and are fixed; we did not re-randomize. The publicly sourced government/government-related corpus is a particular distribution (meeting decks, training material, regulatory presentations) and the result may not transfer to other distributions (commercial pitch decks, conference talks).

Chart grounding is partial. Native PowerPoint chart objects and embedded Excel OLE charts produce exact CHART_* evidence items (titles, categories, series, and values). Screenshot-only charts still depend on vision captions and are not treated as exact numeric evidence. The headline numbers reported above were generated before the TABLE_* and CHART_* kinds were added, so they do not yet reflect whatever change those kinds would make on the 3 of 30 decks in the experimental subset that contain native tables and on the one deck whose embedded .xlsx carries chart data for the service-pack slides.

The pipeline is configured for 39 languages, but only English narration was evaluated; the formal runs skipped audio and video. For other languages, we make no quality claim. Translated narration is re-checked for citation presence and graph consistency; content-level paraphrase preservation after translation is not currently measured.

The language-model fluency rewrite uses gpt-4o-mini as served by OpenAI’s default route on 2026-04-22; the exact dated snapshot was not pinned at run time. The historical manifest also lacks the exact Git commit and rendered-prompt hash. We record the call date and preserve the raw outputs, but a re-run today may not reproduce generation exactly.

The verifier defines a REMOVE verdict but the current decision logic does not emit it. In the reported runs the REMOVE count is zero. The verdict surface for unsupported claims in our evaluated setting is effectively PASS and REWRITE.

The gate routed every evaluated slide to review, and no controlled study measured review time or downstream narration quality. Model cost is not recoverable because token usage was not recorded. The evidence establishes narration-stage execution and observable controls, not end-to-end AV evaluation, labor savings, or cost savings.

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